LESSON 7.6

# Determine Distance on the Coordinate Plane

**Set B · Version A** · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

REMEMBER

## Determine Distance on the Coordinate Plane

**Distance on the Coordinate Plane. Formula: Distance = |x1 − x2| (horizontal) or |y1 − y2| (vertical)** Distance is how far apart two points are, and it is never negative. When two points share a row or a column, you can find the distance using their coordinates.

1. Same quadrant / Same sign: Subtract absolute values of differing coordinates.
2. Different quadrants / Opposite signs: Add absolute values of differing coordinates.
3. Distance is never negative; it is zero when both points coincide

**Words:** **Distance *(Distancia)*** **Absolute value *(Valor absoluto)*** **Horizontal distance *(Distancia horizontal)*** **Vertical distance *(Distancia vertical)*** **Coordinate plane *(Plano cartesiano)***

**1** **Circle the letter of the best answer. Show how you know.**

What is the distance between (-4, 1) and (3, 1)?

(A) -7 units
(B) 1 unit
(C) 7 units
(D) 12 units

**HINT** The y's match (both 1), so it's a horizontal gap between x = -4 and x = 3.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

In which quadrant is the point (-4, 5)?

(A) Quadrant I
(B) Quadrant III
(C) Quadrant II
(D) Quadrant IV

SHOW YOUR WORK

**3** **Circle the letter of the best answer. Show how you know.**

Which point is located on the x-axis?

(A) (-4, -4)
(B) (0, 4)
(C) (4, 4)
(D) (4, 0)

SHOW YOUR WORK

**4** **Circle the letter of the best answer. Show how you know.**

A point has a positive x-coordinate and a negative y-coordinate. Which quadrant is it in?

(A) Quadrant I
(B) Quadrant II
(C) Quadrant IV
(D) Quadrant III

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

A drone starts at position (-4, 0) on a grid and needs to deliver a package to (3, 0), then fly to another delivery at (3, -5). It can only fly horizontally or vertically. What is the horizontal distance from (−4, 0) to (3, 0)?

(A) 1 unit
(B) −1 unit
(C) 7 units
(D) 12 units

SHOW YOUR WORK

**6** **Circle the letter of the best answer. Show how you know.**

A drone starts at position (-4, 0) on a grid and needs to deliver a package to (3, 0), then fly to another delivery at (3, -5). It can only fly horizontally or vertically. What is the vertical distance from (3, 0) to (3, −5)?

(A) 3 units
(B) 5 units
(C) −5 units
(D) 8 units

SHOW YOUR WORK

EXPLAIN YOUR THINKING

**How did you use coordinates and absolute value to find the distance between two points in the same row or column?**

SENTENCE STARTER The points ___ and ___ share the same ___ -coordinate. To find the distance, I found |___| + |___| = ___, which equals ___ units.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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# Determine Distance on the Coordinate Plane

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**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

## PART 1 Understand the idea

models and meaning

**1** **Write the letter of the matching item on each line.**

Match each pair of city landmarks to the number of blocks between them.

_____ Subway (1, 6) to Tower (8, 6)

_____ Market (-3, -1) to Stadium (-3, 7)

_____ Bank (2, 3) to Library (2, 9)

_____ Café (-2, 1) to Gym (-2, -4)

(A) 7 blocks

(B) 8 blocks

(C) 6 blocks

(D) 5 blocks

**2** **Write the name of the correct group on each line.**

Sort each block-distance problem by whether you SUBTRACT (same side of zero) or ADD absolute values (opposite sides of zero).

**Groups:** **Same side of zero — subtract** **Opposite sides of zero — add absolute values**

_____ School (2, 3) to Library (2, 9)

_____ Subway (1, 6) to Tower (8, 6)

_____ Park (-4, 5) to City Hall (3, 5)

_____ Café (-2, 1) to Gym (-2, -4)

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

A drone starts at position (-4, 0) on a grid and needs to deliver a package to (3, 0), then fly to another delivery at (3, -5). It can only fly horizontally or vertically. What is the total flight path?

(A) 2 units
(B) 7 units
(C) 12 units
(D) 35 units

SHOW YOUR WORK

**4** **Circle the letter of the best answer. Show how you know.**

The Bank is at (2, 3) and the Library is at (2, 9). How many blocks apart are they?

(A) 3 blocks
(B) 6 blocks
(C) 11 blocks
(D) 12 blocks

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

The Park is at (-4, 5) and City Hall is at (3, 5). How many blocks apart are they?

(A) 1 block
(B) 5 blocks
(C) 7 blocks
(D) 12 blocks

SHOW YOUR WORK

**6** **Circle the letter of the best answer. Show how you know.**

The Café is at (-2, 1) and the Gym is at (-2, -4). How many blocks apart are they?

(A) 2 blocks
(B) 3 blocks
(C) 5 blocks
(D) 6 blocks

SHOW YOUR WORK

EXPLAIN YOUR THINKING

**How did you use coordinates and absolute value to find the distance between two points in the same row or column?**

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